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Poverty, Inequality, and Fertility: The Contribution of Demographic Change to Global Poverty Reduction

Frank-Borge Wietzke

Global poverty has fallen dramatically over the past decades. In many developing countries, this transformation was accompanied by rapid improvements in demographic outcomes, such as falling child mortality and fertility. Yet, recent theorizing and empirical research into the causes of global poverty reduction has mostly omitted demographic factors. This paper aims to fill this gap by testing for effects of demographic variables on poverty. Using time series data for 140 countries, we document a strong effect of lagged fertility on country-specific poverty rates. This effect is robust across several specifications and data sets. It appears to be stronger in countries with larger fertility differentials, in the early transition stages. The proposed mechanism behind this result is a "Kuznets curve-type" expansion of fertility inequality at the onset of the demographic transition. We conclude by calling for a stronger inclusion of demographic variables in the distribution-sensitive analysis of global poverty.

Introduction

By recent estimates, the world's poverty headcount as calculated at the standard income cut off of PPP$ 1.90 per day has been reduced by over one billion people since 1990, despite continued population growth in many developing countries.[1] Much of this was dominated by East Asia (and mostly China). However, improvements have also been registered in regions with less impressive growth performance, such as South Asia, Latin America, and even sub-Saharan Africa (World Bank 2016a, see also Edward and Sumner 2015). These advances have been hailed as one of the greatest development achievements of our time.[2]

To this point, most of the literature about the drivers of global poverty reduction has focused on economic processes. Dollar and Kraay (2002) influentially argued that faster economic growth in the last decades of the past

Frank-Borge Wietzke, Institut Barcelona d'Estudis Internacionals, 08005 Barcelona Spain. fbwietzke@ibei.org.

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century was often associated with faster movements out of poverty in the developing world. This was summarized in the claim that "growth is good for the poor" and led to the promotion of liberal policies like free trade, or economic and financial deregulation. However, other authors questioned whether poverty reduction is determined by economic growth alone, highlighting instead changes in domestic distributional outcomes as an important mediating mechanism. While this literature mostly focused on the simple arithmetic relationships that connect poverty, inequality, and growth at aggregate level, there has been recognition that more attention needs to be paid to institutional and policy contexts that determine how effectively countries are able to mitigate existing or emerging socioeconomic inequalities that can slow down the transformation of growth into poverty reduction (Bourguignon 2004; Ravallion 2006, 2012; Klasen and Nowak-Lehmann Danzinger 2009).

A question that is absent from recent debates about global poverty concerns the role of demography. Beginning with the 1994 Cairo International Conference on Population and Development (ICPD), most global development initiatives stopped considering population policies as overarching priorities in their own right and instead began to focus on more specific elements of the population agenda, particularly women's empowerment and sexual and reproductive health rights (McIntosh and Finkle 1995; DeJong 2000). This shift to a large degree was motivated by a legitimate desire to move beyond earlier alarmist warnings about population growth and development propagated by neo-Malthusian perspectives like Paul Ehrlich's (1968) *Population Bomb*, the disruptive legacies of coercive birth control policies, as well as by analytical concerns about the potential endogeneity of fertility to poverty and other socioeconomic variables (McNicoll 1997; Dasgupta 2000; Ahlburg and Cassen 2008). However, it has also led to a tendency to exclude population almost entirely from mainstream research on global poverty. For instance, while a widely cited study by Eastwood and Lipton (1999) suggested that lagged birth rates had a significant effect on aggregate poverty rates in the 1980s, the authors also observed (p. 1) that most theorizing about the causes of poverty reduction was "strangely … developed without demographic variables." This situation has not changed significantly since the publication of Eastwood and Lipton's study, with few attempts to incorporate demographic variables directly in the analysis of global poverty dynamics.[3]

The non-consideration of demographic factors is an important omission, given considerable evidence that economic outcomes, demographic variables, and socioeconomic inequality are often closely interrelated. At cross-national level, demographic change has been shown to be at least correlated with, and potentially responsible for, accelerated growth in developing regions like Asia (Bloom and Williamson 1998; Bloom, Canning, and Sevilla 2003). Within countries, countless studies have found

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associations between demographic outcomes, socioeconomic inequalities, and poverty, with considerably higher risks of poverty often observed among lower-status groups with large numbers of children and dependents (Westoff and Cross 2006; Joshi and Schultz 2007; Banerjee and Duflo 2008; Ezeh, Mberu, and Emina 2009; Mberu and Reed 2014; Canning, Sangeeta, and Yazbeck 2015; Dribe et al. 2017). In both cases, demographic variables emerge as closely related to a country's economic potential and its level of domestic inequality—two of the primary explanatory variables typically emphasized in extant literature on global poverty.

This paper aims to fill the gap by reintroducing demographic variables into the analysis of country-specific poverty trends. Building on the aforementioned study by Eastwood and Lipton (1999), it focuses primarily on the distributional effects of fertility on poverty. This is caused by delayed fertility transitions among lower-status groups, which can worsen inequality and result in reduced rates of poverty reduction, especially during the early stages of the demographic transition. We innovate relative to Eastwood and Lipton with the help of new longitudinal data to account more fully for interactions between fertility, inequality, and poverty over time. We also introduce a framework for empirical estimation of the distributional effect of fertility in the (not-uncommon) case where information on fertility differentials within a society is not available. This approach uses more readily accessible data on average fertility rates and builds on the theory of a "demographic Kuznets curve," which was first described in this journal by Eloundou-Enyegue, Giroux, and Tenikue (2017). According to this theory, fertility inequalities increase in the early stages of the demographic transition, when higher-status groups reduce their fertility faster than the poor. They then taper off at later stages of the transition when new reproductive behaviors become more widely disseminated in the population, leading to convergence of fertility rates between the poor and non-poor. Combining these insights of Eloundou-Enyegue, Giroux, and Tenikue with those of Eastwood and Lipton, we predict that this Kuznets-type pattern should translate into an inverted-U shape relationship between the proportion of the population in poverty and average fertility (plotted on an inverted scale to trace countries through their transition from high- to low birth rates). The underlying mechanism is an initial worsening in inequality and poverty in the early stages of the demographic transition, when fertility rates between the poor and non-poor diverge, and a reduction in inequality and poverty in later stages of the transition, when fertility rates between the poor and non-poor converge.

Our empirical analysis broadly supports these propositions. Using data on within-country differences in fertility rates from the Demographic and Health Surveys (DHS), we first document strong signs of the postulated Kuznets-type relationship between average fertility and fertility differentials between the poor and non-poor across countries at different stages of

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the demographic transition. This pattern is less distinct when status differences between the poor and non-poor are defined by schooling, which is consistent with earlier evidence that fertility differentials tend to be particularly "sticky" along educational lines (see, e.g., Bongaarts 2003; Rios-Neto et al. 2018). These results translate remarkably well into the analysis of the relationship between average fertility and the population share in poverty, where we find robust support for the expected inverted-U shaped effect. Moreover, the results are again less marked in countries with higher education inequality.

The article proceeds as follows. The next section reviews in more detail the conceptual framework of economic poverty analysis and discusses the different pathways through which fertility decline can contribute to observed poverty rates. This is followed by an outline of the theory and evidence for the "demographic Kuznets curve" that underpins our prediction of an inverted-U shaped relationship between fertility and poverty. The fifth section presents our data, econometric results, and robustness tests. We conclude by arguing for a stronger inclusion of demographic variables in the analysis of and response to poverty. However, rather than focusing on earlier neo-Malthusian concerns with population growth, this new approach should highlight the underlying distributional processes that jointly determine fertility inequalities and economic deprivation.

Theory and Existing Evidence

Recent economic research on poverty typically conceptualizes the determination of a country's poverty rate as part of a broader three-way association between poverty, growth, and inequality (commonly referred to as the "poverty–growth–inequality triangle"; see Bourguignon 2004). This proposition is based on the simple arithmetic relationship that connects poverty and growth with inequality as the key mediating variable: Poverty, defined in absolute terms, will decrease in the presence of growth, if income gains are registered evenly across the population, such as in a "rising tide lifts all boats" scenario. However, progress in poverty alleviation will be slower if high inequalities limit the "trickle-down" of the effects of growth to the poorest. By implication, poverty reduction will be faster when growth is accompanied by simultaneous reductions in inequality.

This three-way association is often operationalized through variants of a time-series poverty–growth regression of the following form:

$$\begin{aligned}\text{Log Poverty}_{it} = \beta_1 \ \log \ \text{Mean income}_{it} + \beta_2 \log \text{Gini}_{it} \\ + \beta_3 \log \text{Mean income}_{it} * \log \text{Gini}_{it} + \varepsilon_{it} \qquad (1)\end{aligned}$$

where the natural logarithm of the proportion of the population living in poverty ("poverty headcount") in country i at time t is predicted by the

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logarithm of a country's mean income, and where the income Gini typically serves as the measure of inequality. The equation also includes an interaction term between the last two variables to capture the fact that the poor in high-inequality countries tend to benefit less from the gains of economic growth. The Gini index and interaction term jointly aim to capture the influence of domestic inequalities on poverty that are of major concern to more recent distribution-sensitive approaches to poverty analysis (Bourguignon 2004; Ravallion 2006, 2016).

Introducing demographic variables

Poverty–growth regressions of the format described above have served the literature well as a framework for estimating the impacts of growth-promoting and redistributive economic policies on poverty. However, they have also come under increased criticism from new approaches that call for a more multidimensional analysis of human well-being (see, e.g., Sen 1985; Kakwani and Silber 2013). While most of this literature has been concerned with the development of new poverty indices that incorporate additional information on non-monetary dimensions of people's well-being, such as education or health (see, e.g., Duclos, Sahn, and Younger 2006; Alkire, Roche, and Vaz 2017), there is also growing recognition that multidimensionality can matter in a causal sense: Typically, the various dimensions of poverty will interact in perpetuating deprivations over time. Because some of these influences are only imperfectly captured by a single measure like income or consumption, the case is made to also consider other non-monetary variables in the analysis of poverty (Grusky and Kanbur 2006; Thorbecke 2013).

To this point, demographic variables have not featured systematically in the multidimensional analysis of poverty.[4] However, it is possible to envisage several pathways that can connect population trends to poverty. The following discussion focuses on the same channels identified in the aforementioned earlier study by Eastwood and Lipton (1999). They are also summarized in Figure 1.

Under the first channel, population trends affect poverty indirectly, through the influence of decreasing fertility rates on economic growth (see top half of Figure 1). The underlying mechanisms are closely related to the notion of the demographic dividend (Birdsall, Kelley, and Sinding 2001; Bloom, Canning, and Sevilla 2003; Canning, Sangeeta, and Yazbeck 2015). The first demographic dividend arises when falling birth rates in the early- to middling stages of the demographic transition lead to a temporal expansion of the share of the population of working age, followed by an increase in female labor force participation, as women spend less time bearing and bringing up children. Provided the right economic conditions and policies are in place, the resulting rise in output and employment can also lead to in a range of follow-on effects in the form of higher domestic savings and

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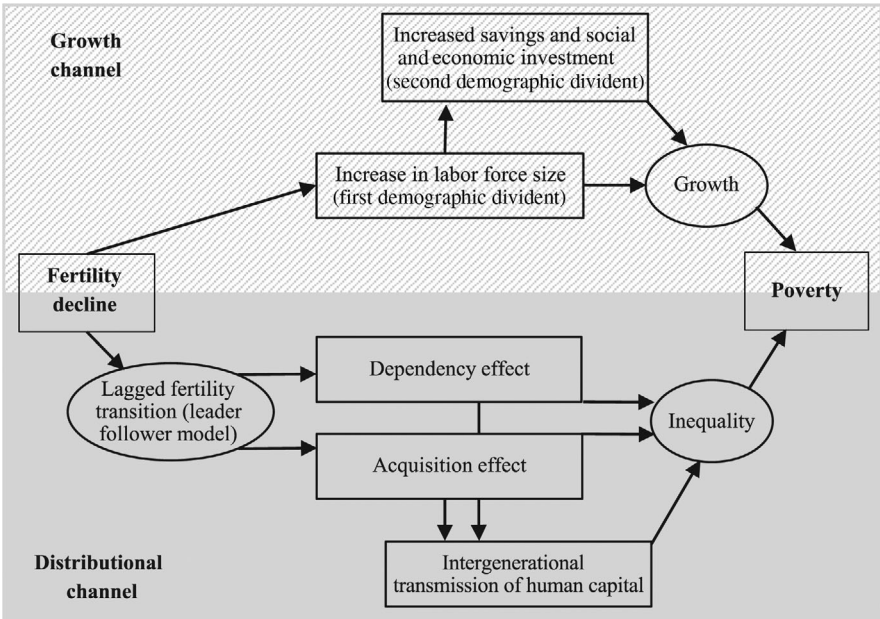
FIGURE 1 Causal pathways from fertility decline to poverty



human and physical capital investments (second demographic dividend; see, e.g., Bloom, Canning, and Sevilla 2003).

Indirect evidence in support of the growth channel has been found in the context of the "East Asian Growth Miracle" (Bloom and Williamson 1998), where fertility decline and subsequent changes in the age structure of the population have been associated with considerable improvements in economic output and poverty (World Bank 2016b; Edward and Sumner 2015), as well as by Eastwood and Lipton's (1999) own analysis, which found a positive effect of fertility reduction on GDP growth and per capita consumption. Further research has provided support for some of the underlying transmission mechanisms, such as the increase in female labor force participation (Bloom et al. 2009), or improvements in education and other social indicators (Lee, Mason, and Miller 2000; Bloom et al. 2007). By contrast, the delayed onset of the fertility transition in sub-Saharan Africa has been identified as one of the factors contributing to the continent's comparatively weaker development performance (Bloom and Sachs 1998; Eastwood and Lipton 2011; Canning, Sangeeta, and Yazbeck 2015).

While there are legitimate questions whether demographic change can be considered as a factor that is fully exogenous to economic growth (Becker and Lewis 1973; Ahlburg and Cassen 2008), there are indications that some of these effects are causal. Historical literature suggests that key demographic events, such as the decline in mortality rates, often preceded growth in economic output and some of its common correlates like

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urbanization (Galor and Weil 2000; Dyson 2001, 2010; Cervellanti and Sunde 2015). Research based on more recent data also suggests that progress in demographic outcomes is often only loosely related to economic development. For example, the well-known "Preston curve" suggests that mortality rates are falling relatively uniformly across countries, regardless of their average level of income (Preston 1975).[5] Although the Preston curve relationship requires some exogenous improvement in health technology and knowledge, these changes can occur even in least developed countries through mechanisms like trade, media and mass education, or development aid (see, e.g., Preston 1975; Canning, Sangeeta, and Yazbeck 2015).

The second effect of fertility on poverty, which is of more central interest to this article, works through a distributional channel (see bottom of Figure 1). It is mostly driven by uneven fertility transitions between the poor and the non-poor. Historical evidence from demographic transitions in today's advanced economies suggests that higher socioeconomic groups often take the lead in the move toward lower fertility rates, followed at a later stage by similar fertility reductions among poorer groups. Fittingly, these processes have been described as the "leader-follower model" of demographic transitions (Bongaarts 2003, see also Stys 1957; Livi-Bacci 1986; Haines 1989, 1992; Skirbekk 2008). Underlying causes include the faster adoption of new birth control methods by better-off groups, due to their superior access to education and health care. Moreover, poorer classes may decide to delay the shift toward lower fertility as a rational response to higher child mortality, the need for additional working family members, and to compensate for incomplete coverage by insurance and social protection systems (Eastwood and Lipton 2001, 214; Ravallion 2016, 367).

The resulting differences in fertility rates between the poor and non-poor can increase inequality and reduce the rate of poverty reduction at given levels of average income. The underlying mechanisms are described at the bottom of Figure 1. The first is a dependency effect, as larger and poorer households need to divide scarce resources among more dependents. The second is an acquisition effect, driven by increased constraints on the ability of poorer parents to save or engage in income-generating activities after the birth of another child (Eastwood and Lipton 1999). When an intergenerational perspective is added, these mechanisms can also explain the "stickiness" of poverty over time, as children from larger and poorer households are less likely to receive the education and the health and social care that they would require to compete on equal terms with children from smaller and better-off households in later phases of their life (Ravallion 2016, 354f).

Empirical support for these effects of uneven fertility reductions on poverty is provided at aggregate level by Eastwood and Lipton in their own test of the distributional effect of fertility on poverty. Taking average rates of fertility decline in Asian economies as a benchmark, their results suggest that a reduction in (lagged) birth rates by 5 per 1000 inhabitants would

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have led to a reduction in the national poverty headcount from close to 19 percent to 12.6 percent in the average developing country. Their analysis controls for mean income and the possible interaction between fertility and income through the growth channel. The authors estimate that about half of the observed effect of falling birth rates can be attributed to the distributional channel. Eastwood and Lipton's results are also consistent with earlier work from India, which found a small but significant association between population growth and the local poverty incidence in 13 Indian States (van de Walle 1985; see also Chelliah and Sudarshan 1999), while contradicting results by Ahlburg (1996), who found no significant relationship between population growth and poverty, but only on a small cross-sectional sample of 22 countries. Nonetheless, also Eastwood and Lipton's results are based on a relatively limited cross section of 59 countries, which makes it worthwhile to update their analysis below with a larger sample and a more explicit over time perspective.

Numerous studies have also provided supporting microlevel evidence of fertility differentials between the poor and non-poor that underpin the distributional channel. Analysis of over 600 household surveys from developing countries by Olinto et al. (2013) finds that key predictors of higher fertility, like agricultural employment or larger female education gaps, tend to be concentrated among the extreme poor. This is consistent with the notion of a delayed fertility transition that drives fertility differentials between the rich and the poor. The authors also find a disproportionate share of children in poorer households, which hints at a dependency effect of fertility on poverty. Earlier studies based on cross-sectional and panel household surveys similarly document a higher poverty risk and more persistent instances of poverty among larger families with greater dependency ratios (Krishnaji 1984; Bielicki 1986; Glewwe 1990; Gaiha and Deolalikar 1993; Coulombe and McKay, 1994; Lloyd 1994; Banerjee and Duflo 2008).

While the above evidence is mostly based on observational data, there are again indications that these effects are causal. Joshi and Schultz (2007) employ the famous Matlab family planning quasi-experiment in Bangladesh to study the impact of population policies on economic well-being. They find a fertility decline of around 15 percent in villages that were randomly selected into the program, followed by subsequent improvements in key predictors of poverty, such as household earnings and assets, the intergenerational transmission of schooling and health, and general improvements in women's health and use of preventive medicine. Drawing on data from India, Rosenzweig and Wolpin (1980) use the unexpected increase in the number of children per household that is caused by the birth of twins to identify a negative effect of larger family size on children's school enrolment. Mussa (2014) finds that increased fertility raises the risk of consumption poverty in Malawi, even though households with more children are less likely to describe themselves as poor in subjective poverty as-

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sessments. This study uses son preferences as an instrument to deal with the possibility that fertility levels are endogenous to households' poverty status.

Turning to the acquisition effect, Ebenstein (2009) and Cruces and Galiani (2007) find a negative influence of fertility on labor force participation which disproportionally affects women. However, several other studies find no or mixed effects of another child on labor force participation, as poorer households in particular often respond to the increased financial needs of feeding another child by supplying additional labor (Agüero and Marks 2011; Priebe 2011; Longwe, Smits, and de Jong 2013). All of these contributions tackle the potential endogeneity of fertility choices by exploiting exogenous sources of variation, such as fertility shocks or the gender composition of born children.

Differences between Demographic and Economic Determinants of Poverty over Time

A primary innovation of this article relative to the earlier work by Eastwood and Lipton is the introduction of an explicit longitudinal perspective in the analysis. In this context, it is of particular interest to discuss how the distributional processes discussed in the previous section influence poverty over time.

In the conventional income-based approach to poverty analysis, longer-term distributional trends typically come to the fore in the context of the Kuznets curve hypothesis (Kuznets 1955; see also Bourguignon 2004; Ravallion 2016, 396). First developed by economist Simon Kuznets on the basis of data from industrialized societies, this hypothesis predicts an inverted-U shaped relationship between inequality and average income: Inequality should first increase in earlier stages of development where income growth is often concentrated in more dynamic industrial or urban sectors. It then decreases once productivity and wage gains spread within the population. By implication, the Kuznets curve regularity would point to slower rates of poverty reduction in the early phases of economic transformation, when the poor are less likely to benefit from growth, and accelerated movements out of poverty at later stages when income gains are shared more widely among the population.

The problem is that the Kuznets curve relationship, as originally defined in the space of incomes, is not typically supported by very consistent evidence. In particular, studies that have traced income inequalities within countries over time on the basis of more recent data than the sources used by Kuznets find little or no evidence of the expected concave relationship between income and inequality (see, e.g., Banerjee and Duflo 2003; Piketty 2006). Possible reasons include structural factors, such as continued technological change (Piketty 2006), differences in the sectoral composition of growth (Rodrik 2016), as well as context-specific influences,

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such as a country's redistributive policies and institutional environment (Bourguignon 2004). For the analyst, this complicates predictions about the long-term rate of poverty reduction, as well as policy recommendations on how the poverty impact of growth can be increased. In the words of Bourguignon (2004: 11), knowledge of the "identity linking poverty reduction, growth and distribution is certainly not sufficient to establish the optimal mix of growth and distribution oriented policies in a development strategy." Typically, more detailed information about a country's institutional and policy context is required to arrive at reliable recommendations for local poverty alleviation strategies (see also Ravallion 2006).

A central argument of this article is that a focus on fertility may point to more predictable patterns. This claim builds on earlier work by Eloundou-Enyegue, Giroux, and Tenikue (2017) in this journal who posit that fertility transitions under the previously described leader–follower model should result in a process that resembles a Kuznets curve: Differences in fertility between the poor and non-poor should first increase in the early stages of the demographic transition, when changes in reproductive behavior are mostly adopted by forerunners in better-off socioeconomic groups, but fertility remains high among the poor. The level of fertility inequality should then taper off again at later stages, when new family planning behaviors and techniques are disseminated more widely among the population.

Empirical support for this "demographic Kuznets curve" has been provided by Eloundou-Enyegue, Giroux, and Tenikue for 40 sub-Saharan countries as well as from other studies that compared fertility inequalities across a number of developed and developing economies (see, e.g., Skirbekk 2008; Bongaarts 2003). However, it is important to recognize that also in this case the evidence base is far from perfect. Research by demographers often suggests that the convergence in fertility rates between the poor and nonpoor, which would give the relationship its predicted inverted-U shape, does not always materialize. This is particularly the case in countries with strong and persistent educational inequalities, which can slow down the spread of new reproductive behaviors among the population (Hull and Hull 1977; Bongaarts 2003; Rios-Neto et al. 2018), as well as in societies where the poor face high barriers of entry into skilled jobs or quality health care (see, e.g., Ferreira and Paes de Barros, 1999; Lanjouw and Ravallion, 1999; Ravallion 2016, 351ff.). Perhaps more importantly, so far only relatively few countries report disaggregated data on fertility behaviors for sufficiently long-time periods to allow tracing the evolution of fertility differentials over the full duration of the demographic transition. To this day, most of the contemporary evidence on the demographic Kuznets curve is either based on cross-sectional comparisons or on data series that are too short to document the full extent of initial divergence or eventual convergence of fertility rates between the poor and nonpoor (Eloundou-Enyegue, Giroux, and Tenikue 2017).

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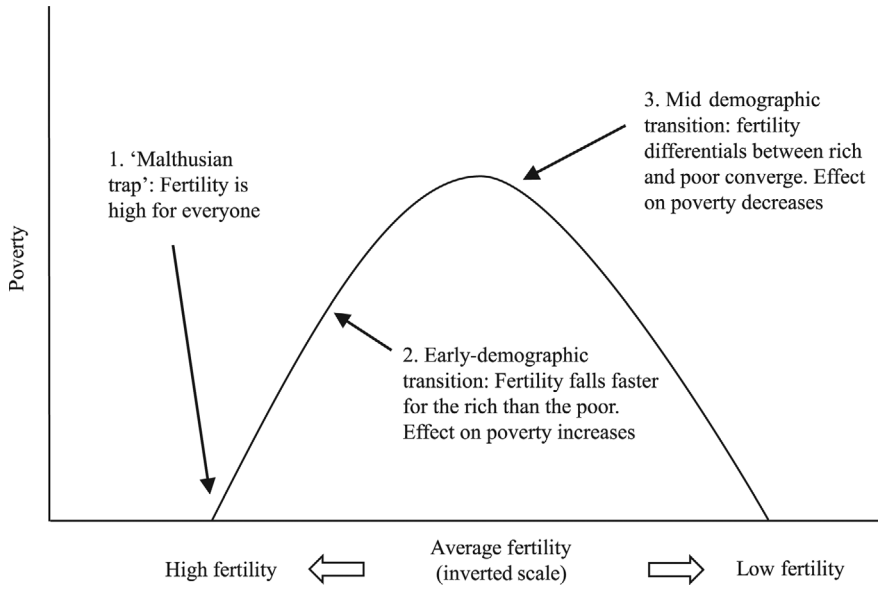
FIGURE 2 The distributional effect of fertility on poverty



Notwithstanding this latter concern, we argue that it should be possible to infer evidence of a possible Kuznets-type effect of fertility on poverty, even on the basis of more limited information. In the following analysis, we focus on average total fertility rates, for which longer-run country-level data series are typically more readily available from harmonized sources like the UN population statistics. The underlying intuition of our argument is summarized in Figure 2 and outlined below.

The horizontal axis in Figure 2 captures average fertility rates on an inverted scale to line up countries as they move through the demographic transition from high to low fertility. The vertical axis describes the expected effect of fertility on the poverty headcount. Following Eastwood and Lipton's (1999) original work, we further assume that the analysis controls for income and interactions between falling birth rates and economic development, to account for a possible effect of fertility decline on poverty through the growth channel outlined at the top of Figure 1. Figure 2, in other words, only focuses on the distributional effect of fertility on poverty as previously discussed in our review of the work of Eastwood and Lipton.

At the beginning (left-hand side) of the graph, societies are still caught in the "Malthusian trap." Fertility rates are uniformly high for the poor and nonpoor, leading to low rates of income growth for both groups. The distributional effect of fertility on the poverty headcount under these conditions should be neutral, especially when we account separately for the negative effects of high fertility on economic development as mentioned above.

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In the early transition phase, birth rates start falling for better-off households but remain stable for poorer populations. This worsens distributional outcomes by raising the fertility differential between lower and higher status groups and—proportionally speaking—shifts the burden of high birth rates onto poorer populations. Fertility under these conditions should cease to have a neutral effect on poverty, again controlling for average income and other possible interactions between fertility decline and economic conditions through the growth channel. Accordingly, the expected effect of fertility on poverty through the distributional channel increases.

In the middling stages of the demographic transition, new reproductive behaviors and birth control techniques become more widely disseminated among the population, eventually leading to convergence in fertility rates between the poor and the nonpoor. Throughout this process, the effect of fertility on poverty through the distributional channel declines until it becomes neutral again at the point where differences in fertility between the poor and the non-poor are no longer discernible. This decreasing effect leads to the inverted-U-shape of the relationship between poverty and fertility in Figure 2.

Is There Evidence of a Demographic Kuznets Curve?

The notion of a demographic Kuznets curve is sufficiently central to our argument about the distributional effect of fertility on poverty to merit further analysis. In the following, we turn to information from the DHS. These provide survey-based fertility estimates for adult women (aged 15–49) and other socioeconomic attributes that allow tracing differences in reproductive outcomes across different segments of the population. As noted before, the primary disadvantage of data sources like the DHS is their limited coverage in space and over time. For the present analysis, we have data for 64 developing nations but the country-specific data series are typically too short to allow tracing fertility differentials through the full demographic transition.

The graphs in Figure 3 respond to these limitations by plotting the data as they would appear in conventional cross-country poverty–growth regressions, as country-year observations (some countries have multiple entries, depending on the number of surveys available. Country codes are not reported for all observations due to space limitations). Panel 3(a) plots fertility differentials between the poorest and the wealthiest 60 percent of the population measured by an index of asset wealth (Booysen et al. 2008). This is followed by fertility comparisons between urban and rural areas (Panel 3b), and between uneducated and educated women (Panel 3c). Average fertility rates (horizontal axis) are again reported on an inverted scale to enable us to line up countries on the path from high to low fertility. Additional

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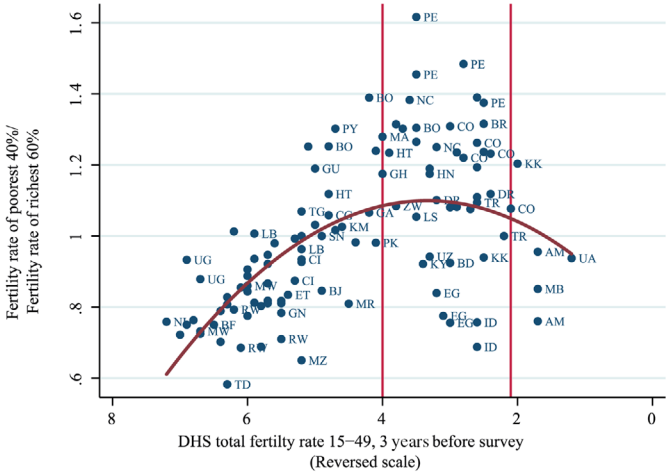
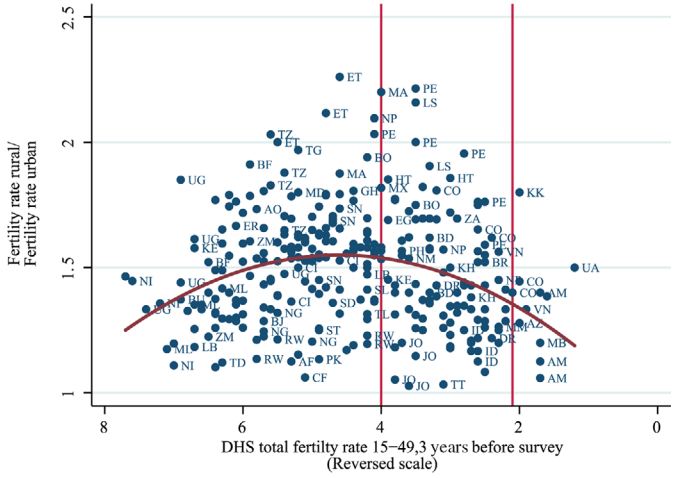
FIGURE 3 Demographic Kuznets curve, descriptive evidence

Panel 3a Fertility differentials, by asset ownership



The fitted line omits observations with fertility differentials larger than 1.4. The coefficient of the fitted line is 0.2 (linear term) and −0.03 (squared−term), p = 0.000 for both coefficients with a 95% CI, R^2=0.41.

Panel 3b Fertility differentials, urban - rural



The fitted line omits observations with fertility differentials larger than 2.2. The coefficient of the fitted line is 0.29 (linear term) and −.03 (squared−term), p = 0.000 for both coefficients with a 95% CI, R^2=0.09.

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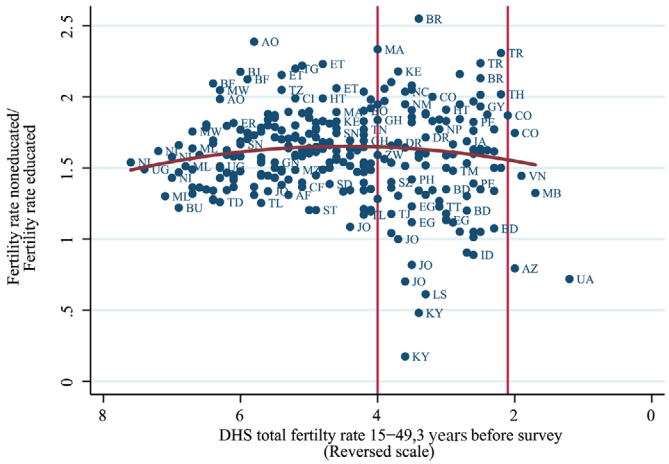
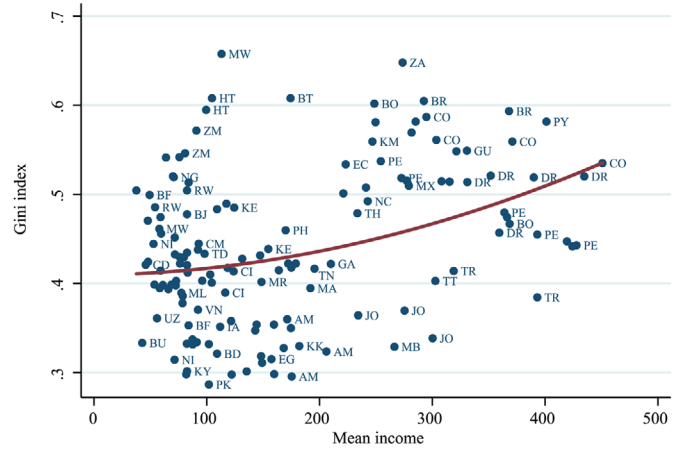
FIGURE 3 (continued)

Panel 3c Fertility differentials, non-educated vs educated



The graph omits two outliers from Albania and Yemen. The fitted line omits cases with fertility differentials below 0.9. The coefficient of the fitted line is 0.15 (linear term, p = 0.069) and −0.02 (squared−term, p = 0.063), with a 95% CI, R^2=.012.

Panel 3d Gini index over mean income



The graph omits one outlier from Ukraine. The coefficient of the fitted line is 0.00004 (linear term, p = 0.21) and 4.53e−07 (squared−term, p = 0.98) with a 95% CI, R^2=.097.

SOURCE: DHS surveys. Labels for overlapping observations omitted.

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vertical lines are included at an average fertility rate of 4 and at the replacement level of 2.1 births per woman, respectively, to distinguish mid-demographic transition countries from early- and post-transition countries (see, e.g., Bongaarts 2003). For comparison purposes, we also look for evidence of a Kuznets-type relationship in the more conventional space of incomes, using data on income inequality (measured by the Gini) and average income (Panel 3d) from the World Bank's World Development Indicators for the same set of countries.

We treat the breakdown over the asset index as our preferred specification, because it has the closest conceptual links to income poverty and inequality (Panel 3a). This graph provides strong support for the proposed demographic Kuznets curve relationship. Despite considerable variation in the data, the regression line reveals the expected inverted U-shape. This relationship is highly significant, even though we exclude extreme cases like Peru with fertility differentials above 1.4 (see notes in Panel 3a for coefficients and p values). Fertility differentials peak in countries in the middle of the demographic transition. This is followed by signs of convergence in countries where fertility has reached or is approaching replacement level, albeit with decreasing numbers of observations as we approach lower fertility rates. The results are broadly in line with the aforementioned study of Eloundou-Enyegue, Giroux, and Tenikue (2017) as well as Skirbekk (2008).[6]

Panel 3b shows similar patterns when we evaluate fertility levels between rural and urban areas. However, the Kuznets-type relationship is much less distinct when we plot fertility differentials between educated and uneducated mothers (Panel 3c). This is consistent with aforementioned literature, which often highlights educational differences as a particularly "sticky" form of inequality that can considerably delay fertility transitions among poorer populations (Hull and Hull 1977; Bongaarts 2003; Skirbekk 2008; Rios-Neto et al. 2018). Nonetheless, also in this panel fertility differentials are much lower in a number of mid- and late-transition countries, such as Jordan (JO) and Kyrgyzstan (KY). Finally, in all graphs the concave relationship is much stronger than in the comparison case where income inequality (the Gini) is plotted against average income (Panel 3d). This supports our hypothesis above that Kuznets-type patterns of inequality are easier to document for demographic variables than in the space of incomes, where the theory of the Kuznets curve was first developed.

Despite these relatively clear results, the cross-sectional nature of our analysis also leads to potential limitations. This applies in particular to possible region-specific biases in our data. For instance, observations in the pre-transition stage are dominated by African countries, while cases with the highest fertility differentials are in Latin America—a possible reflection of traditionally high levels of socioeconomic inequality in that region. By contrast, Eastern European and Central Asian countries with traditionally

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low inequality and birth rates dominate the right-hand side of the graph, where the fitted regression line returns to its origin. Robustness tests below will account for these regional differences to make sure that they do not drive our results.

Estimating the Effect of Average Fertility on Poverty

The remainder of this article tests our hypothesis that it should be possible to observe a Kuznets curve–mediated distributional effect of fertility on poverty, even on the basis of simpler data that do not record explicitly the level of fertility inequality within societies. To do so, we return to a standard poverty–growth regression as in Equation (1). To this, we add a term for more readily available average fertility rates, as well as a quadratic term for fertility to account for the predicted inverted U-shaped effect of average fertility on poverty. As our discussion of Figure 2 suggests, if the Kuznets-type relationship also drives fertility patterns outside of countries with DHS data, it should be visible in increasing distributional effects of fertility on poverty in countries in the early stages of the demographic transition (high average fertility), when fertility differentials between the poor and non-poor widen. By contrast, it should decrease in countries further advanced in the demographic transition (lower average fertility), when fertility differentials between the poor and nonpoor are starting to converge.

The data that we use to test these assumptions consist of country-level observations on poverty, inequality, and income (all survey-based and converted into 2011 PPP terms) from the World Bank's PovcalNet,[7] as well as average total fertility rates for women aged 15–49 years from the World Bank's World Development Indicators and the UN's population statistics. Information is available for 140 developing countries and a period from the late 1970s to 2016, but with gaps in the data for poverty and economic outcomes for some of the countries. To deal with this problem, and the fact that national estimates of fertility and other demographic outcomes are often imputed between census and survey years, we collapse all available information into country-specific five-year averages, starting in 1975 (robustness tests below indicate that our results are not affected by this manipulation). After accounting for missing observations and excluding high income countries, this leaves us with an unbalanced panel of close to 500 observations for the 140 countries, which span the period from the late 1970s to 2016 (see Tables A1a and A1b in the online Appendix for descriptive statistics).[8]

Equation (2) describes our expanded regression model. Average fertility and fertility squared enter with a 10-year lag to deal with endogeneity concerns and account for possible delays in the effect of higher fertility on households' poverty risk. However, we experiment below with alternative lag structures, so we denote the time lag here in more general notation as

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t-k. The dependent variable is the log of the country-specific poverty headcount at the international "extreme" PPP$ 1.90 poverty line.

$$\begin{aligned}\text{Log Poverty}_{it1} = {} & \beta_1 \log \text{Mean income}_{it1} + \beta_2 \log \text{Gini}_{it1} \\ & + \beta_3 \log \text{Mean income}_{it1} * \log \text{Gini}_{it1} + \beta_4 \text{Fertility}_{it1-k} \\ & + \beta_5 \text{Fertility}^2_{it1-k} + \beta_6 \text{Fertility}_{it1-k} * \log \text{Mean incomee}_{it1} \\ & + \beta_7 \text{Fertility}^2_{it1-k} * \log \text{Mean incomee}_{it1} \\ & + \beta_8 \emptyset + \beta_9 \delta + \beta_{10} X_{it} + \varepsilon_{it} \end{aligned} \tag{2}$$

Following Eastwood and Lipton's (1999) original analysis and our preceding discussion, our preferred model also controls for contemporary average household consumption (henceforth "income"),[9] and for interaction terms between income and the two lagged fertility variables. These are meant to account for a possible indirect effect of fertility decline on poverty through the growth channel as outlined at the top of Figure 1. As is common, we use income estimates from the same survey that produces the poverty estimates for the dependent variable. However, alternative specifications that used GDP per capita from national accounts produced qualitatively similar results (not reported here).

All estimates further include country- and time-period fixed effects $(\emptyset, \delta)$, respectively, to account for possible time-invariant country-level influences and broader, period-specific cross-country trends. X_{it} is a vector of controls described below as part of our robustness tests. Following convention, we transformed all continuous variables into their natural logs, except for fertility, for which we assume a concave function.[10] Standard errors are clustered at the country level, to account for possible heteroskedasticity and within-panel autocorrelation in our data.

Before we turn to the regression results, it is important to address one remaining practical concern about the proposed estimation. Because we take as our starting point the "standard" econometric specification for poverty–growth regressions from Equation (1), our estimates also include a measure of socioeconomic inequality, namely the Gini index. This raises the question why we would expect to see a separate distributional effect of fertility on poverty, given that levels of inequality in a society are already accounted for.

We consider this objection primarily as a practical problem where the response is closely linked to limitations in the way monetary measures like the income Gini approximate levels of inequality in a society. The first of these concerns the time horizon that that is typically used to construct estimates of income inequality in the type of poverty–growth regressions that also underpin our analysis. Following convention, we use income Ginis from the same surveys that produce poverty estimates for the

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dependent variable (see, e.g., Eastwood and Lipton 1999; Ravallion 2006). These estimates tend to emphasize short-term changes in inequality, such as fluctuations caused by macroeconomic shocks or policy reforms. However, they are likely to misrepresent crucial drivers of fertility inequality like education, mortality, or socioculturally determined fertility preferences, which are typically slower changing (see, e.g., Bongaarts 2003).

Another reason is that income or consumption measures that underpin estimates of the Gini in Equation (1) usually inadequately capture crucial non-monetary determinants of uneven fertility behaviors. For example, it is well known that household consumption and income aggregates that are widely used in the economic analysis of poverty and inequality typically do a particularly poor job in describing key predictors of high fertility, such as mortality, insufficient access to health care and (female) education, or uneven gender relations within households (Grusky and Kanbur 2006; Ravallion 2016). The upshot of both of these observations is that we still expect to see an independent distributional effect of fertility on poverty, even when measures of monetary income inequality like the Gini are included in the estimation. To provide further support for these claims, we experiment below with interaction terms between fertility and different monetary and non-monetary variants of the Gini index that allow testing whether income- and fertility-based dimensions of inequality indeed pull in different directions.

Results

Estimates in Table 1 start from a benchmark model with only income, the Gini, and the interaction between these two variables (column 1) to look for signs of omitted variable bias in the economic predictors when demographic information is not included in the estimation. As expected, income enters with a robust negative coefficient. This initial estimate is reduced by well over a third until we reach the preferred specification with fertility, fertility squared, and the interaction of these two variables with income (column 4). The increase in the R-squared between columns 1 and 4 is comparatively more moderate–it rises by about 0.074 (from 0.743 to 0.817) or 10 percent of the initial value. This suggests that most of the unobserved effect of fertility in the first column was absorbed by omitted variable bias in the model's economic right-hand-side variables. Nonetheless, the results in no way cancel out the role of average income as an independently significant and quantitatively important predictor of poverty. The sign of the coefficient for the Gini fluctuates between negative and positive across specifications and the estimates are never very robust. However, specifications without the interaction term between income and the Gini produced a significant positive effect for inequality, consistent with expectations (not reported).

Turning to our main variables of interest, the linear term for lagged fertility enters with the expected positive sign. This coefficient increases more

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TABLE 1 The effect of fertility on poverty

	(1) Without Fertility	(2) Without Fertility-squared	(3) With Fertility-squared	(4) Preferred model	(5) With interaction with stage of demographic transition	(6) With interaction with income Gini	(7) With interaction with education Gini
Mean income	−2.887**	−2.220*	−1.958*	−1.750*	−2.481**	0.436	−3.358**
	(1.196)	(1.193)	(1.117)	(0.973)	(1.048)	(1.266)	(1.528)
Gini	−0.291	0.453	0.731	−1.029	−0.532	2.371	−2.890*
	(1.630)	(1.591)	(1.495)	(1.096)	(1.273)	(1.584)	(1.680)
Interaction Mean income × Gini	0.412	0.239	0.196	0.526**	0.434*	−0.247	0.935***
	(0.307)	(0.305)	(0.286)	(0.217)	(0.247)	(0.325)	(0.345)
Fertility lagged 10 years		**0.199*****	**0.601*****	**4.367*****	**0.960*****	**0.284**	**6.282*****
		(0.043)	**(0.101)**	**(0.707)**	**(0.241)**	**(1.393)**	**(1.647)**
Fertility-squared lagged 10 years			**−0.041*****	**−0.430*****		**0.024**	**−0.641*****
			(0.009)	**(0.074)**		**(0.156)**	**(0.179)**
Interaction Fertility lagged 10 years × mean income				−0.695***	−0.084**	−0.439***	−0.695***
				(0.128)	(0.035)	(0.096)	(0.178)
Interaction Fertility-squared lagged 10 years × mean income				0.070***		0.045***	0.070***
				(0.013)		(0.011)	(0.019)
Interaction Fertility × Early-Transition both lagged 10 years					−0.430***		
					(0.086)		
Interaction Fertility × Posttransition both lagged 10 years					−0.424**		
					(0.174)		
Early-transition lagged 10 years					1.672***		
					(0.322)		
Posttransition lagged 10 years					0.611*		
					(0.339)		

/...

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TABLE 1 (continued)

	(1) Without Fertility	(2) Without Fertility-squared	(3) With Fertility-squared	(4) Preferred model	(5) With interaction with stage of demographic transition	(6) With interaction with income Gini	(7) With interaction with education Gini
Interaction Fertility lagged 10 years × Gini lagged 10 years						0.712**	
						(0.344)	
Interaction Fertility-squared lagged 10 years × Gini lagged 10 years						−0.082**	
						(0.038)	
Gini lagged 10 years						−1.307*	
						(0.751)	
Interaction Fertility lagged 10 years × Education Gini lagged 10 years							−0.525**
							(0.216)
Interaction Fertility-squared lagged 10 years × education Gini lagged 10 years							0.058**
							(0.025)
Education Gini lagged 10 years							1.179**
							(0.556)
Observations	498	498	498	498	498	420	381
R-squared	0.743	0.766	0.785	0.817	0.807	0.785	0.849
Countries	140	140	140	140	140	120	101

NOTES: All estimates include a constant term and country and five year fixed effects. Continuous variables except fertility and fertility-squared have been transformed into natural logs. Robust standard errors in parentheses. ***$p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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than threefold when we add the quadratic term, which is itself negative and significant, confirming the expected inverted-U shaped relationship between fertility and poverty. Both coefficients increase further in size when we include interaction terms with income to account more fully for the growth effect of fertility on poverty. We treat this as our preferred specification because it distinguishes most explicitly between the distributional and the growth effect of fertility on poverty outlined in Figure 1. However, the signs of the interaction terms between fertility, fertility-squared, and income suggest that these effects are less marked in higher income countries. This is also consistent with sensitivity tests presented in the Appendix of this article, which indicate that the effect of fertility is not significant at higher poverty lines (Table A3 in the Appendix).

Estimates in column 5 add more structure to the model, by replacing the term for squared fertility with dummies and corresponding interactions for early- and post-transition countries.[11] The significant negative effects of the two interaction terms support the notion that the distributional effect of fertility is highest in countries in the middling stages of the demographic transition, which serve here as the reference category. Again, this is consistent with the idea of an inverted U-shaped relationship between fertility and poverty.

Although our estimates do not account directly for fertility differentials between the poor and non-poor, we can get a sense of the extent of interaction between fertility and socioeconomic inequalities by including controls for the product of fertility and (lagged) measures of inequality. The next two models do so by adding interactions between the two fertility terms and, respectively, the income Gini (column 6) and the education Gini (column 7). These specifications produce very different results. In the model with the interaction with the income Gini, the two fertility terms are now individually not significant. This hints at co-linearities that support the concern discussed above that the inclusion of the control for income inequality may pick up part of the distributional effects of uneven fertility transitions that we are interested in here. Nonetheless, the signs on the coefficients of the two interaction terms suggest that the inverted-U shaped relationship between fertility and poverty is mostly driven by countries with higher levels of income inequality. This result is broadly consistent with the leader-follower model that underpins our argument.

By contrast, the two fertility variables remain individually significant in the model with the education Gini. Moreover, the signs on the interaction terms suggest that the concave effect of fertility is flattened in countries with larger educational differences. This result is broadly consistent with the less distinct curve for educational fertility differentials in Figure 2c. Again, it supports prior evidence from demographic literature, which suggests that high and persistent levels of educational inequality can significantly delay the convergence of fertility differentials in the later stages of the

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demographic transition (Bongaarts 2003; Rios-Neto, Miranda-Ribeiro, and Miranda-Ribeiro 2018). Taken together, these last two results also back our earlier claim that a single measure of inequality like the income Gini will typically not suffice to track the complex and multiple processes that drive interactions between poverty and inequality across different dimensions of well-being.

Predicted effects

To give an indication of the magnitude of the estimated effects, we present predicted values of poverty at different levels of fertility in Figure 4 Panels a–d. Panel 4a reports results for the entire estimation sample, followed by specific graphs for countries that are representative of different stages of the demographic transitions (we use Zambia as an example of an early transition country and Bangladesh and Georgia as respective examples of mid- and late-transition countries). To facilitate interpretation, predicted poverty rates have been reconverted from the logarithmic scale into their original percentage values. Note, however, that this increases the estimates' confidence intervals, as we are reintroducing variation that was suppressed when poverty was predicted in logs. We somewhat arbitrarily fix mean incomes and Ginis at values for the period from 2000 to 2005, but similar results were obtained for alternative time periods. Average fertility rates on the horizontal axis are again presented on an inverted scale.

In the full sample, the expected poverty headcount peaks at around 32 percent, when a country reaches an average fertility rate between 5.5 and 5 (Panel 4a). The concave function that underpins these estimates leads to the somewhat counterintuitive result that predicted levels of poverty initially increase with falling fertility. We highlight again that these should not be interpreted as absolute effects. They are *conditional* estimates of the distributional effect of fertility on poverty, net of income, and the interaction between fertility and income through the growth channel. The effect of fertility on poverty then decreases again once lower average fertility rates are attained, as the theory and the significant negative coefficient for squared fertility would suggest. It is also encouraging that the reversal in the expected effect of fertility decline happens relatively early in the demographic transition, suggesting that also poorer and higher fertility countries should be able to reap benefits of falling fertility relatively soon within their development process.

The country-specific conditional effect estimates are to a large degree driven by the differences in national average incomes at which we fixed values for our predictions. Still, broadly similar patterns emerge between the three cases. In a pre-transition country like Zambia, where fertility never falls significantly below six children per woman in the time period studied, poverty at given incomes is still expected to rise to around 60 percent, before

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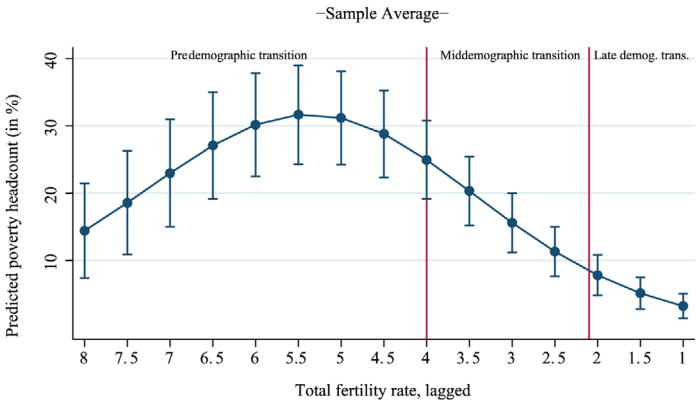
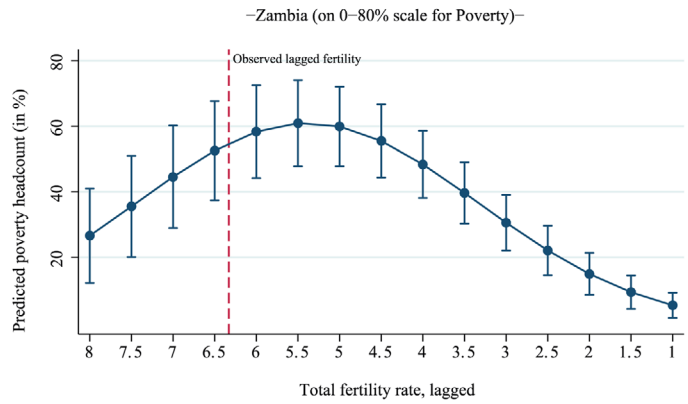
FIGURE 4 Predicted effects on poverty

Panel 4a Conditional marginal effect of fertility on poverty



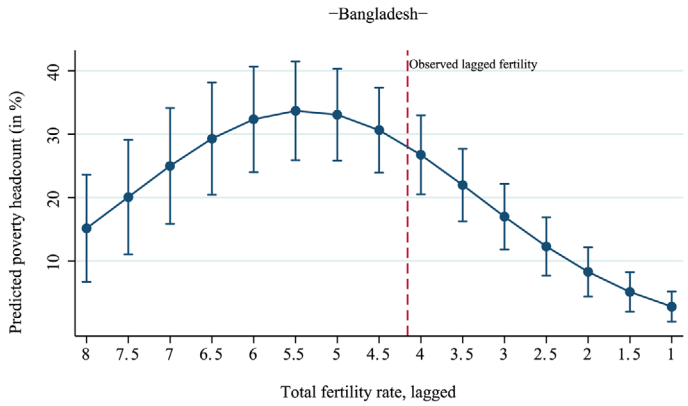
Predicted values of poverty have been transformed back into the variable's original percentage scale.
Estimates with 95% Confidence Interval

Panel 4b Conditional marginal effect of fertility on poverty



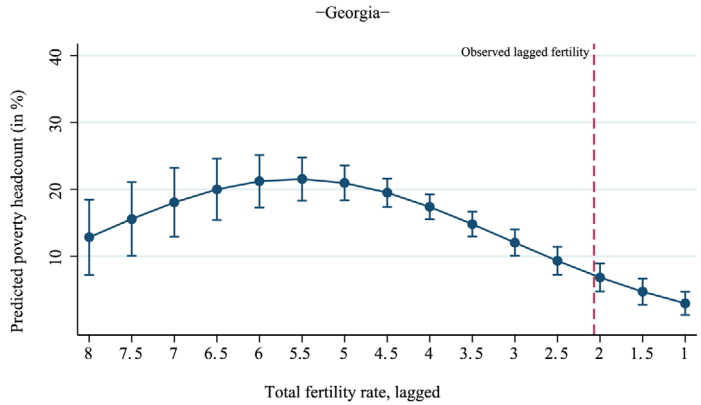
Predicted values of poverty have been transformed back into the variable's original percentage scale.
Estimates with 95% Confidence Interval

Panel 4c Conditional marginal effect of fertility on poverty



Predicted values of poverty have been transformed back into the variable's original percentage scale.
Estimates with 95% Confidence Interval

Panel 4d Conditional marginal effect of fertility on poverty



Predicted values of poverty have been transformed back into the variable's original percentage scale.
Estimates with 95% Confidence Interval

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the benefits of falling fertility will be shared more widely in the population (Panel 4b, note that the scale for poverty ranges from 0 to 80 percent). A country like Bangladesh, which enters the early to middling stages of the demographic transition during the time period analyzed (lagged fertility of 4.16), would see an increase in its poverty rate to about 32 percent, if its fertility rate were to return to 5.5 (Panel 4c). A late-transition country like Georgia (average lagged fertility just below replacement level at 2.07) would see its poverty rate treble from approximately 7–21 percent (Panel 4d).

Robustness tests

Table 2 presents a series of robustness and specification tests, starting out from the preferred model in column 4 of Table 1 (see Table A2 in the Appendix for full results). In the first column of Table 2, we turn to the original "raw" data to demonstrate that we obtain similar results when we do not average variables over five year periods. In column 2, we return to DHS data. As noted above, fertility rates in national statistics are often imputed between census and survey years, so there is a legitimate concern that this may affect our previous estimates. The DHS surveys help deal with this problem, because all the demographic information they provide is directly observed at the household level. The results confirm the expected concave relationship between poverty and lagged fertility.[12] However, possibly due to the much smaller sample size, we do not find significant effects for fertility when we include the two interaction terms between mean income and fertility and fertility squared. The interaction terms themselves also mostly seem to add noise. They are not individually significant (column 3).

The DHS also allow testing directly for the effect of observed fertility differentials on poverty. We do so by replacing the squared term with the measure of fertility differentials between the poorest and the wealthiest 60 percent of the population on the asset index that is reported in Figure 2a. The specification also includes an interaction between this variable and a dummy that identifies observations where fertility differentials are larger than 1, to distinguish countries where the poor actually have more children than the non-poor (column 3). The fertility differentials variable and the interaction term both enter with a positive sign, which supports the claim that the observed influence of fertility on poverty is transmitted through the distributional channel, linked to delayed fertility transitions among the poor.

The remaining estimates return to our original data set. Results in columns 5 and 6 exploit the longitudinal nature of the data more fully. Column 5 includes lagged values for the dependent variable, income, the Gini coefficient (each lagged 10 years), and additional lagged values of fertility and fertility squared (both lagged 15 years). Column 6 reports results from a dynamic panel model that uses lagged values of the dependent variable

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TABLE 2 Robustness tests. Dependent variable PPP$ 1.90 headcount

Data sets	(1) Large sample, no 5 year averages	(2) DHS data	(3) DHS data	(4) DHS data with observed fertility differentials	(5) Large sample, lagged-dependent variable	(6) Large sample, dynamic panel model	(7) Large sample, with additional controls	(8) Large sample, with regional interactions	(9) Large sample, without LAC and ECA
Fertility lagged 10 years	**4.310*****	**0.480*****	**−0.091**	**−0.406****	**5.183*****	**4.781*****	**3.081*****	**4.756*****	**2.688*****
	(0.727)	**(0.134)**	**(1.138)**	**(0.194)**	**(0.898)**	**(1.065)**	**(0.698)**	**(0.901)**	**(0.895)**
Fertility-squared lagged 10 years	**−0.416*****	**−0.034*****	**−0.029**		**−0.545*****	**−0.482*****	**−0.373*****	**−0.594*****	**−0.320*****
	(0.082)	**(0.012)**	**(0.121)**		**(0.104)**	**(0.114)**	**(0.080)**	**(0.109)**	**(0.094)**
Interaction Fertility lagged 10 years × mean income	−0.669***		0.062	0.087**	−0.948***	−0.835***	−0.582***	−0.803***	−0.504***
	(0.131)		(0.223)	(0.037)	(0.161)	(0.186)	(0.129)	(0.144)	(0.159)
Interaction Fertility-squared lagged 10 years × mean income	0.067***		0.004		0.103***	0.085***	0.071***	0.096***	0.061***
	(0.015)		(0.025)		(0.020)	(0.021)	(0.019)	(0.018)	(0.018)
Fertility differential				0.397*					
				(0.223)					
Interaction Fertility differential × Fertility differential >1 (dummy)				0.114***					
				(0.028)					
Fertility differential >1 (dummy)				−0.706***					
				(0.133)					
With lagged-dependent variables	No	No	No	No	Yes	Yes	No	No	No
With additional controls	No	No	No	No	No	No	Yes	Yes	Yes
With regional interactions	No	No	No	No	No	No	No	Yes	No
Observations	1,049	147	147	112	243	130	268	268	148
R-squared	0.823	0.870	0.876	0.938	0.872	0.910	0.890	0.897	0.902
Number of countries	139	64	64	52	102	59	90	90	53

Abbreviations: DHS, Demographic and Health Surveys; ECA, Eastern European and Central Asia; LCA, Latin American and Caribbean countries.
NOTES: All estimates include a constant term, controls for income and the Gini, the product of mean income and the Gini, and country and five-year fixed effects. Estimates in columns 5 and 6 include lagged values of the dependent variable (lagged 10 years) and fertility variables (lagged 15 years). Estimates in columns 7–9 also control for the education Gini, old-age dependency ratio, under-5 mortality rate, population proportion in working age (15–64), population share in labor force, female labor force participation rate, and female education. Continuous variables except fertility and fertility-squared have been transformed into natural logs. Robust standard errors in parentheses. ***$p < 0.01$, ** $p < 0.05$, *$p < 0.1$.

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as instruments (Arellano and Bond 1991). Both of these tests account more fully for possible autocorrelation and endogeneity problems, such as due to past interactions between fertility, income, and poverty. The results for fertility are still significant. However, the introduction of additional lagged variables considerably reduces the sample size, so we do not treat these models as our preferred specifications.

Estimates in column 7 introduce additional time-variant controls that were not included in the more parsimonious specifications of Table 1, due to concerns about missing values and sample size. These include (again) the educational Gini coefficient, as well as controls for old-age dependency ratios and the under-5 mortality rate. The latter controls help explore possible changes in the dependency effect of rising fertility on the poor, for example, because of labor substitution by older household members or more surviving children. Other controls added to the model account for influences that would be more consistent with explanations that connect fertility to poverty through the growth channel. These include the population proportion in the working age (15–64), the population share in the labor force, the female labor force participation rate, female education, and the gross domestic savings rate. We show below that the first two of these variables are robustly predicted by lagged fertility, so adding them significantly raises the bar for our estimates of the effect of lagged fertility. Nonetheless, our results are still significant.

Finally, we control for regional influences to account for the strong geographic cross-sectional variation in fertility differentials visible in our analysis of DHS surveys in Figure 3. Estimates in column 8 add interactions between the two fertility variables and region-specific dummies to the model (using Eastern Europe and Central Asia as the omitted category). Detailed results in Table A2 (Annex) point to few significant region-specific differences. However, a substantively important result is a less distinct concave effect of fertility on poverty in sub-Saharan Africa. This is broadly consistent with prior evidence that the convergence of fertility differentials appears to happen much more slowly in this region (see, e.g., Bongaarts 2003). Estimates in column 9 drop Latin American and Caribbean and Eastern and European and Central Asian countries, to account for the fact that the concave slope of the regression line in Panel 3a appeared to be driven by observations from these regions. Again, the coefficients of the two fertility terms remain significant.

Transmission mechanisms

Estimates in columns 1–3 of Table 3 explore in more depth transmission mechanisms behind the postulated distributional channel from fertility to poverty. We focus on the dependency effect where the availability of data is better than for the acquisition effect. In column 1, we show that we obtain

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TABLE 3 Transmission mechanisms

Dependent variable	(1) Poverty	(2) Average of children <5 stunted, wasting, anemic	(3) Survival in primary school	(4) Population proportion in working age	(5) Population proportion in labor force	(6) Female labor force participation rate	(7) Government primary education spending (% of GDP)	(8) Domestic savings (% of GDP)
Mean income	−1.311	−0.674*	−0.574*	−0.132***	−0.229*	−0.113	−0.402	2.505
	(0.967)	(0.342)	(0.327)	(0.036)	(0.127)	(0.151)	(0.810)	(1.528)
Gini	−1.661	−0.834**	−0.474	−0.128**	−0.270	−0.081	−0.237	2.322
	(1.065)	(0.371)	(0.491)	(0.062)	(0.176)	(0.258)	(1.136)	(2.188)
Young-age dependency lagged 5 years	**0.442*** **							
	(0.055)							
Young-age dependency-squared lagged 5 years	**−0.003*** **							
	(0.000)							
Interaction Young-age dependency lagged 5 years × mean income	−0.066***							
	(0.009)							
Interaction Young-age dependent-squared lagged 5 years × mean income	0.0004***							
	(0.0001)							
Fertility lagged 10 years		**0.581** **	**−0.748*** **	**−0.204*** **	**−0.261*** **	**−0.276**	**−0.313**	**1.614**
		(0.257)	**(0.283)**	**(0.048)**	**(0.099)**	**(0.260)**	**(0.646)**	**(1.042)**
Fertility-squared lagged 10 years		**−0.076** **	**0.089** **	**0.018*** **	**0.026** **	**0.024**	**0.043**	**−0.258** **
		(0.029)	**(0.035)**	**(0.005)**	**(0.012)**	**(0.023)**	**(0.075)**	**(0.126)**
Interaction Fertility lagged 10 years × mean income		−0.087*	0.133**	0.027***	0.030*	0.031	0.085	−0.301
		(0.052)	(0.053)	(0.009)	(0.018)	(0.037)	(0.118)	(0.201)
Interaction Fertility-squared lagged 10 years × mean income		0.013**	−0.016**	−0.003***	−0.004*	−0.003	−0.011	0.050**
		(0.006)	(0.007)	(0.001)	(0.002)	(0.004)	(0.013)	(0.024)
Observations	500	308	392	490	445	445	394	405
R-squared	0.834	0.734	0.212	0.880	0.572	0.254	0.130	0.135
Number of countries	138	113	123	138	135	135	123	121

NOTES: All estimates include a constant term, the interaction of mean income and the Gini, and country and five-year fixed effects. Continuous variables except fertility and young-age dependency ratio have been transformed into natural logs. The lag structure for fertility and fertility-squared in column (2) is five years. Robust standard errors in parentheses. ***p < 0.01, ** p < 0.05, * p < 0.1.

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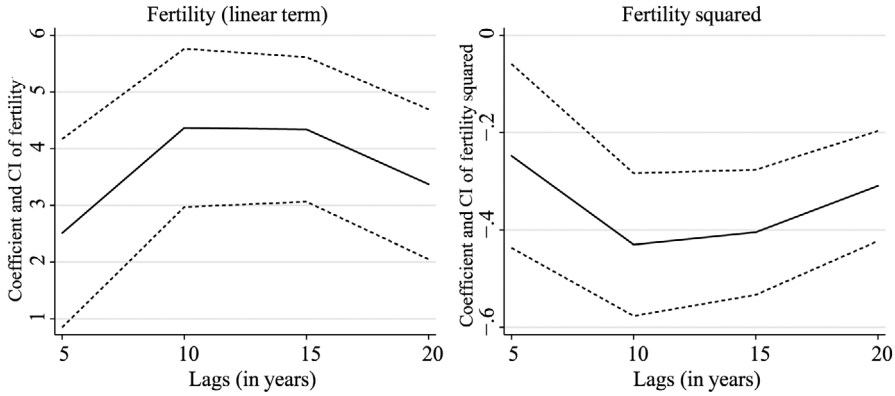
FIGURE 5 Effects of fertility at different time lags



NOTE: Solid lines track the coefficient estimates. Dashed lines track the confidence intervals.

similar results when we regress poverty directly on young age household dependency ratios. Column 2 indicates that higher lagged fertility increases the risk of stunting, wasting, or anemia in children under 5,[13] and that it decreases the percentage of children surviving primary school to the last grade (column 3). Both results would be consistent with a situation where households with more dependents are forced to reduce their investment in the welfare of each individual child.

By contrast, we obtain much more mixed results for variables that would point to a possible transmission through the growth channel. Lagged fertility rates have a positive effect on the population share in the working age and in the labor force (columns 4 and 5). In particular, the former is typically viewed as a structural precondition for a demographic dividend, so we controlled for these variables in our robustness tests in Table 2. However, results are much more mixed for other outcomes that typically need to improve if a country is to turn its *potential* for a demographic dividend into the *actual attainment* of that dividend, including female labor force participation, government education spending, or domestic savings (columns 6–8). In our sample, lagged fertility rates have no significant effect on any of these variables. Again, this provides support for the claim that the observed effect of fertility on poverty is less likely to be driven by the growth channel.

Finally, we can get a sense of underlying transmission mechanisms by considering the effect of fertility at alternative lag structures (Figure 5). The size of the coefficient of fertility peaks between a lag of 10 and 15 years, but it drops visibly at longer lag structures. Individuals in developing countries born at time t–k would normally begin to enter the labor market after a period of 15 years, so these results again provide more support for the idea that the influence of fertility on poverty is driven by a dependency effect, rather than through the growth channel.

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Discussion and Conclusion

We conclude by discussing the implications of our findings for the wider debate about population and poverty. Over the past decades, this topic has elicited widely different and often conflicting responses. Neo-Malthusians, who dominated the discourse on population and development for much of the 1960s and 1970s, and whose views are becoming again more influential in recent debates at the intersection of demography, climate, and conflict, traditionally highlight the potential negative effects of population growth on economic well-being and poverty. This is typically accompanied by arguments about the pressures expanding populations can put on natural resources and linked to calls for more proactive forms of population control (see, e.g., Ehrlich 1968; Nett and Rüttinger 2016, for a critique see Dasgupta 2000).

At least since the 1994 Cairo International Conference on Population and Development, a much more nuanced consensus has emerged. The conference itself moved away from the often-alarmist claims of the neo-Malthusians and emphasized instead the role of wider socioeconomic influences as joint determinants of poverty and fertility. Global policy initiatives since Cairo consequently prioritize a more inclusive approach, built around the promotion of reproductive rights, women's empowerment, and maternal and child health (McIntosh and Finkle 1995; DeJong 2000).

The move away from an explicit focus on population dynamics was also compounded by analytical concerns about the difficulty to reliably identify the effects of fertility on poverty (McNicoll 1997; Dasgupta 2000). Since around 2000, research on global poverty and economic development has been conducted almost without attention to demographic variables (for exceptions see Birdsall, Kelley, and Sinding 2001; Ahlburg and Cassen 2008). Remaining studies have concentrated almost exclusively on the microlevel, with a focus on natural or policy experiments to address concerns about endogeneity and possible context-specific variability in the relationship between fertility and poverty (see above and Dasgupta 2000; Joshi and Schultz 2007). This trend has only changed recently with a number of flagship reports by leading aid agencies like the World Bank and the United Nations that highlight again the role of demographic processes in economic and social development (see, e.g., World Bank 2016b; Canning, Sangeeta, and Yazbeck 2015; UNDP 2016).

This article has followed this more recent literature by reintroducing population in the analysis of global poverty. Building on earlier cross-national analysis by Eastwood and Lipton (1999), we have shown that lagged fertility has a sizeable and significant effect on country-specific poverty rates that survives a battery of robustness and specification tests. While it could be argued that this turn to the aggregate level merely trades off the analytical rigor of more narrowly defined microstudies for the greater

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external validity of cross-national data, we see the advantage of this shift primarily in the ability to trace broader regularities in the interaction of demographic variables and poverty that are more easily overlooked at the microlevel. Specifically, we expand on earlier work by Eloundou-Enyegue, Giroux, and Tenikue (2017) in this journal to argue that fertility connects to poverty through longer-term changes in fertility inequality that broadly follow the same concave pattern identified earlier by the economist Kuznets (1955) for the space of incomes. Subsequent analysis indicates that this process translates into an inverted-U shaped relationship between average fertility and poverty. The result is the somewhat counterintuitive prediction that the net distributional effect of fertility reductions at given levels of economic development can be an increase in poverty during the early stages of the demographic transition, when the fertility differential between the poor and nonpoor tends to expand. More unambiguously positive effects of fertility decline on poverty then emerge during the middle and later stages of the demographic transition, when new family planning practices become more widely disseminated among the population.

Our findings have a number of broader implications for the debate about population and poverty. The fact that the estimated influence of fertility on poverty works primarily through a distributional channel points to a potential way forward to reconcile concerns about population dynamics with the rights- and justice-based approaches that have come to dominate the field since the mid-1990s. More precisely, our results suggest that the current focus on reproductive and women's rights in global development initiatives can be an important mechanism to accelerate the conversion of economic development and growth into poverty reduction, if it is accompanied by effective and well-targeted interventions that offset pre-existing differences in fertility determinants between poorer and better-off groups.

In a similar vein, our findings provide further justification for the move toward a more multidimensional conceptualization and analysis of poverty. Indeed, in addition to demonstrating that demographic processes matter for the understanding of global poverty dynamics, our findings point to other dimensions that should be included in the analysis. This applies in particular to educational inequalities, which emerge as an important factor that can delay the shift toward universally lower fertility and resulting faster rates of poverty reduction.

The primary caveat of our analysis is the relatively limited evidence on which our underlying arguments about the demographic Kuznets curve are based. As with the initial formulation of a concave relationship between inequality and income formulated in the 1950s by Kuznets, the results of this study still rely on preliminary and largely cross-sectional evidence. Future analysis, based on more and better longitudinal information from developing countries, may well point to qualifications of our argument, similar to the gradual adjustments in the evidence base for the original

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income-based Kuznets curve. As such, we view our analysis as a first foray into the data, to be followed by more in-depth research, as the availability of longer-term longitudinal information from developing regions continues to improve.

Notes

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1 All dollar values in this paper are reported as daily per capita incomes, converted in 2011 purchasing power parity (PPP) terms. Global poverty estimates differ with PPP conversion rates and poverty lines (see, e.g., Edward and Sumner, 2015).

2 See, e.g., *The Economist* June 1, 2013, "The world's next great leap forward: Towards the end of poverty."

3 The literature often uses population projections for poverty forecasting. However, demographic variables are not typically considered in the analysis of *past* poverty trends. (For exceptions, see Ferreira and Paes de Barros, 1999; Birdsall et al., 2001 Ahlburg and Cassen, 1996; Sinding, 2009.)

4 However, multidimensional poverty analysis often considers correlates of demographic outcomes, such as education, child health, or gender equality; see, e.g., Alkire et al. (2017).

5 Note, however, that absolute mortality rates are still much higher in lower income regions than in the developed world.

6 There are also a large number of countries in the predemographic dividend stage, where richer groups have higher fertility rates than the poor. While this finding is consistent with earlier evidence of class-based fertility differentials in today's advanced economies (see, e.g., Haines, 1989), it may be an artifact in our data. In particular, the asset index that is used to distinguish poorer and richer households may identify higher socioeconomic status positions for families with more members, due to economies of scale in the accumulation of household assets.

7 http://iresearch.worldbank.org/PovcalNet/home.aspx, last accessed December 2018.

8 Appendices are available at the supporting information tab at wileyonlinelibrary.com/journal/pdr.

9 We typically have consumption data, except for some Eastern European, Central Asian, and Latin American countries, which rely on household income. We used the data type with the longest period available for each country, but ensured that data sources are consistent within countries.

10 Specifications without logs produced comparatively stronger and more robust estimates for the demographic variables, so we consider our approach here as a more conservative solution.

11 Country-specific values of the dummies vary in time (lagged 10 years), so the coefficients can be estimated, even with country fixed effects included.

12 DHS surveys are not always available for the same years and time periods as our estimates on poverty and average household income. For this reason, the lag structure in the demographic variable ranges from 7 to 13 years.

13 The latter two estimates use a 5-year lag for dependency ratios and fertility because we focus here on outcomes for younger children. However, similar effects were obtained with 10 year lags.

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